Curriculum Vitae – Whitney Leach, Ph.D.

DeSales University
2755 Station Ave. Center Valley, 18034 USA
whitney.leach@desales.edu

a. Education

2015-19 *University of North Carolina at Charlotte.* Charlotte, NC. Ph.D., Environmental, Ecological, & Evolutionary Biology. Advisor: Adam Reitzel, Ph.D.

2009-13 *University of North Carolina at Charlotte.* Charlotte, NC. B.S., Biology. Advisor: Stanley Schneider, Ph.D.

b. Academic/Professional Appointments

2025- **Assistant Professor**. DeSales University. Biological Sciences.

2024-25 **Postdoctoral Research Associate**. Lehigh University. Advisor: Mike Layden, Ph.D.

2020-24 **Postdoctoral Research Associate**. Stowers Institute for Medical Research. Advisor: Matt Gibson, Ph.D.

2015-19 **Graduate Research Assistant**. *University of North Carolina at Charlotte*. Advisor: Adam Reitzel, Ph.D.

2016-17 **Graduate Teaching Assistant**. *University of North Carolina at Charlotte.* Animal Physiology Lab. Summer 2016, Fall 2016, Summer 2017.

2016 **Graduate Teaching Assistant**. *University of North Carolina at Charlotte.* General Biology II Lab Spring 2016.

2010-13 **Undergraduate Research Assistant**. *University of North Carolina at Charlotte*. Advisor: Stan Schneider, Ph.D.

c. Publications

1. **Leach, W.B.,** Babonis, L., Juliano, C.E., Nakanishi N., Schnitzler, C.E., Steinmetz, P.R., Layden, M.L. Discoveries and innovations in cnidarian biology at Cnidofest 2024. EvoDevo. 2025 Jun 16; 16(1). DOI: 10.1186/s13227-025-00247-5 https://doi.org/10.1186/s13227-025-00247-5
2. Ketchum, R.N., Smith, E.G., Toledo, L.M., **Leach, W.B.**, Padillo-Anthemides, N., Baxevanis, A.D., Reitzel, A.M., Ryan, J.F. Rapid speciation in the holopelagic ctenophore *Mnemiopsis* following glacial recession. *bioRxiv 2024.10.10.617593; doi: https://doi.org/10.1101/2024.10.10.617593*
3. Al-Shaer, L., **Leach, W.B.,** Baban, N., Yagodich*, M., Gibson, M.C., Layden, M.J. 2023. Environmental and molecular regulation of asexual reproduction in the sea anemone *Nematostella vectensis. R. Soc. Open Sci.10230152.* https://doi.org/10.1098/rsos.230152
4. Sipprell S.E., Johnson M.B., **Leach W.B.,** Suptela S.R., Marriott I. 2023. Staphylococcus aureus infection induces the production of the neutrophil chemoattractants CXCL1, CXCL2, CXCL3, CXCL5, CCL3, and CCL7 by murine osteoblasts. Infect Immun. 2023:e0001423
5. Johnson M.B., Furr K.H., Suptela S.R., **Leach W.B.**, Marriott I. 2022. Induction of protective interferon-beta responses in murine osteoblasts following *Staphylococcus aureus* infection. Front Microbiol. 2022;13:1066237.
6. **Leach W.B.,** Reitzel A.M. 2020. Decoupling behavioral and transcriptional responses to color in an eyeless cnidarian. *BMC Genomics.* 21:361. https://doi.org/10.1186/s12864-020-6766-y.
7. Ketchum R.N., Smith E.G., DeBiasse M.B., Vaughan G.O., McParland D., **Leach W.B.**, Al-Mansoori N., Ryan J.F., Burt J.A., Reitzel A.M. 2020. Population genomic analyses of the sea urchin *Echinometra* sp. *EZ* across an extreme environmental gradient. *Genome Biology and Evolution*. https://doi.org/10.1093/gbe/evaa150.
8. Sachkova M.Y., Macrander J., Surm J.M., Aharoni R., Menard-Harvey, S.S., Klock A., **Leach W.B.**, Reitzel A.M., Moran Y. 2020. Some Like It Hot: Population specific adaptations in venom production to abiotic stressors in a widely distributed cnidarian. BMC Biology. *Accepted.*
9. **Leach W.B.**, Reitzel A.M. 2019. Transcriptional remodeling upon light remodeling in a model cnidarian: losses and gains in gene expression. *Molecular Ecology.* 28:14 3413-3426. doi.org/10.1111/mec.15163.
10. **Leach W.B.,** Carrier T.J., Reitzel A.M. 2019. Diel patterning in the bacterial community associated with the sea anemone *Nematostella vectensis*. *Ecology and Evolution.* 9:17 9935-9947. doi:10.1002/ece3.5534.
11. **Leach W.B.,** Peres R., Macrander J., Reitzel A.M. 2018. Transcriptome-wide analysis of differential gene expression in response to light:dark cycles in a model cnidarian. *Comparative Biochemistry and Physiology – Part D.* doi.org/10.1016/j.cbd.2018.03.004.

d. Invited Seminars / Lectures

1. Whitney B. Leach, University of Florida's Whitney Marine Lab Friday Seminar Series, 2023. Exploring the bZIP regulatory network of *Nematostella vectensis* circadian clock.

e. Conferences, Symposia and Workshops *(Most Recent)*

1. Jamie Havrilak, Minghe Cheng, **Whitney Leach**, Michael Layden. Utilizing single-cell RNA sequencing to identify neural subtypes in *Nematostella vectensis.* Abstracts of Cnidofest: The Cnidarian Model Systems Meeting 2024 Conference, Bethlehem University, Bethlehem, PA. 2024.
2. Andrea I. Frías Vellón*, **Whitney B. Leach**, Jeff Lange, Matthew C. Gibson. Dark Necessities: Uncovering the effects of light on larval settlement in *Nematostella vectensis.* Abstracts of Society for Integrative and Comparative Biology Conference, Austin, Texas. 2023.
3. **Whitney B. Leach**, Lacey Ellington, Stephanie Nowatarski, Sofia Robb, Matthew C. Gibson. Stella-EP: *Nematostella vectensis* Expression and Phenotype Knowledgebase. Abstracts of Cnidofest: The Cnidarian Model Systems Meeting 2022 Conference, University of California at Davis, Davis, CA. 2022.
4. **Whitney B. Leach,** India I. Gaume*, Matthew C. Gibson. An inquiry into the molecular and behavioral disparities of a sexually dioecious but monomorphic sea anemone. Abstracts of Society for Integrative and Comparative Biology Conference, Phoenix, Arizona. 2022
5. **Whitney B. Leach** and Adam M. Reitzel. The impacts of wavelength on gene expression and behavior in the sea anemone *Nematostella vectensis.* Abstracts of Astrangia Research Workshop, Portsmouth, Rhode Island. 2019.
6. **Whitney B. Leach** and Adam M. Reitzel. Transcriptional remodeling upon light remodeling in a model cnidarian: losses and gains in gene expression. Abstracts of Society for Integrative and Comparative Biology Conference, Tampa, Florida. 2019.
7. **Whitney B. Leach** and Adam M. Reitzel. Transcriptome Dynamics After Light Removal in a Model Cnidarian, *Nematostella vectensis.* Abstracts of Cnidofest: The Cnidarian Model Systems Meeting, 2018 Conference, Whitney Marine Laboratory for Marine Bioscience, St. Augustine, Florida. 2018.
8. Jason Macrander, Maria Sachkova, **Whitney Leach,** Remi Ketchum, Amy Klock*, Yehu Moran, and Adam M. Reitzel. Toxin Assemblage and Prey Community in the Model Sea Anemone *Nematostella vectensis* throughout their natural range. Abstracts of Cnidofest: The Cnidarian Model Systems Meeting 2018 Conference, Whitney Marine Laboratory for Marine Bioscience, St. Augustine, Florida. 2018.
9. Meredith J. Collins*, **Whitney B. Leach**, Adam M. Reitzel. Effects of Chronic and Acute Salinity Stress on the Behavior of a Burrowing Sea Anemone, *Nematostella vectensis.* Abstracts of Cnidofest: The Cnidarian Model Systems Meeting 2018 Conference, Whitney Marine Laboratory for Marine Bioscience, St. Augustine, Florida. 2018.
10. **Whitney B. Leach,** Rafa Peres, Jason Macrander and Adam M. Reitzel. Transcriptomic Changes in Response to a Diel Light-Dark Cycle in *Nematostella vectensis*, an Estuarine Anemone. Abstracts of Society for Integrative and Comparative Biology Conference, San Francisco, California. 2018.
11. **Whitney B. Leach**. 2017. Here Comes the Night: 'Stella Opsins and Behavior – She's got Rhythm. EcoDevoGeno 2.0, Whitney Lab, St. Augustine, FL.
12. **Whitney B. Leach,** Rafa Peres, and Adam M. Reitzel. Diel Lighting Impacts on Behavior and Opsin Expression in a Coastal Cnidarian. Abstracts of Society for Integrative and Comparative Biology Conference, New Orleans, Louisiana. 2017.
13. **Whitney B. Leach**, Adam M. Reitzel. Effects of Light Cycling and Wavelength on the Molecular Components and Circadian Clock of *Nematostella vectensis.* Abstracts of The Association of Southeastern Biologists Conference, Concord, North Carolina. 2016.
14. **Whitney B. Leach**, Adam M. Reitzel. Effects of Light Cycling and Wavelength on Gene Expression and Behavior in a Model Cnidarian. Abstracts of UNCC Graduate Research Symposium. 2016.
15. **Whitney B. Leach.** LEDs and Anemones: Party All Night Long. EcoDevoGeno, *University of North Carolina at Charlotte*, Charlotte, NC. 2016.

* Denotes undergraduate co-author

e. Synergistic Activities/Research Experience

Service and Outreach

2019 DNA Day at Lincoln Charter High School, *Charlotte NC*.
2019 Visiting Scientist Program at Starmount Academy of Excellence, *Charlotte NC*.
2018 Collaboration with Discovery Place, *Charlotte NC*.
2017 Sea Anemone Touch Experience at University Park Creative Arts Elementary School – STEAM Week, *Charlotte NC*.
2016 UNC Charlotte Science and Technology Expo – Biology of Insects, *University of North Carolina at Charlotte.*
2016 Visiting Scientist Day at Carmel Christian High School – NASA Club, *Charlotte NC.*
2010 Joint Undergraduate Mentoring Program, Counselor, *University of North Carolina at Charlotte.*

2010-12 Supplemental Instruction Leader, Principles of Biology I & II, *University of North Carolina at Charlotte.*
2011 Summer Intensive Chemistry Orientation Leader, *University of North Carolina at Charlotte.*

Mentoring

Undergraduate* and Graduate Students Mentored

- Mikayla Lighting* (Investigating mechanisms of sex determination in *Nematostella vectensis*) *Desales University* Class of 2025.
- Andrea I. Frías Vellón* (Effects of light on *Nematostella vectensis* behavior during development; B.S. 2022) *Stowers Institute for Medical Research.*
- Fanning Xie (Functional Exploration of *Nematostella vectensis* Opsins) *Stowers Institute for Medical Research.* Current position: PhD Candidate, Trainor Lab
- India Gaume* (Behavioral and genomic differences between male and female *Nematostella vectensis*; B.S. 2022) *Stowers Institute for Medical Research.*
- Adam Clodfelter* (*Nematostella vectensis* behavioral responses to variable environmental conditions; B.S. 2019) *University of North Carolina at Charlotte*. Current position: PA at Person Memorial Hospital Roxboro, NC in the ER.
- Kendra Shaw* (*Nematostella vectensis* burrowing behavior in response to temperature stress; B.S. 2019) *University of North Carolina at Charlotte*.
- Meredith Collins* (*Nematostella vectensis* burrowing behavior in response to salinity stress; B.S. 2018) *University of North Carolina at Charlotte*. Current position: Research Director at RA Capital Management.

Memberships

2024 *Cnidarian Model Systems Meeting*, Planning Committee Member, 2024
2017- *Society for Integrative and Comparative Biology (SICB)*, Member.
2016-17 *University of North Carolina at Charlotte Association of Biological Sciences Graduate Students (ABSGS)*, Social Chair.
2015-16 *University of North Carolina at Charlotte Association of Biological Sciences Graduate Students (ABSGS)*, Member.

Awards

2018 The Graduate School Summer Fellowship (GSSF). The Graduate School, *University of North Carolina at Charlotte*. Charlotte, NC. Award amount: $6000.00. Awarded for providing support to post candidacy students during the summer.
2018 The Fellowship Application Incentive Program (FAIP). The Graduate School, *University of North Carolina at Charlotte*. Charlotte, NC. Award amount: $1000.00. Awarded for successful application to competitive grants.
2018 Graduate and Professional Student Government Travel Award. The Graduate School, *University of North Carolina at Charlotte*. Charlotte, NC. Award amount: $500.00. Awarded to support travel to present research at a conference.
2017 NSF Supplement: Graduate Student Preparedness (Dear Colleague Letter NSF 16-067). Parent Award: NSF 1536530. Award amount: $34,403. Awarded for education expenses, training and experience in elementary STEM education, production of public ready materials and display for Discovery Place Charlotte, NC.
2017 Graduate and Professional Student Government Travel Award. The Graduate School, *University of North Carolina at Charlotte*. Charlotte, NC. Award amount: $500.00. Awarded to support travel to present research at a conference.
2015-18 Graduate Assistant Support Plan Award. The Graduate School, *University of North Carolina at Charlotte*. Charlotte, NC. Award amount: $11,048 *('17-'18),* $7,986.0 *('16-'17)*, $3,096.00 *('15-'16).* Awarded support for graduate tuition, fees, and health insurance.

Selective Workshops and Training

2019 Stowers Institute Molecular Techniques Training Workshop 2019, *Stowers Institute for Medical Research*, Kansas City, MO – *Technical training in microinjection, CRISPR-Cas9 mediated transgenesis, shRNA synthesis, and electroporation.*
2018 Pasteur Single Cell Workshop 2018, *Institut Pasteur*, Paris, France – *Generation and analysis of single cell transcriptome data from transgenic animals*.
2018 NABI 2018, Providence, Rhode Island – National Alliance for Broader Impacts Summit. *Training for scientists to gain skills in outreach and broader impact to make science more accessible to the public.*
2018 STEM ED, *University of North Carolina at Charlotte*, Charlotte, NC – Teaching Science to Elementary School Learners. *Basic education-based course to provide prospective elementary school teachers with an understanding of the curriculum, methods, and rational for teaching science in grades K-6.*

2016 MEGA 2016, *University of Texas at Austin*, Austin, TX – Methods in Ecological Genomic Analysis. *Foundational training and implementation of tools and bioinformatics programs for TagSeq*.

2016 NC SeaGrant Graduate Training Workshop, New Bern, NC – Science Communication Training.

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